

LIFE CYCLES OF STARS

Reference

See pp. 3 - 7 of Science Aspects Book 4.

Background

There are three possible pathways for stars to follow in their lives. All stars come from a similar start point, but it is their size that ultimately determines their final destiny. Some become black dwarfs, some neutron stars, and some black holes.

Task

Your task is to use the Internet to find useful information that can provide *simple answers* to the following questions. Several useful sources are listed below.

- Good information - probably the *best site* to get information.

<http://www.enchantedlearning.com/subjects/astronomy/stars/lifecycle/giant.shtml>

- Activity on aging photos of stars.

<http://btc.montana.edu/ceres/html/LifeCycle/stars1.html> - activity2

- Photos from Hubble.

<http://hubblesite.org/newscenter/newsdesk/archive/releases/1999/20/>

- Power point - black holes.

<http://imagine.gsfc.nasa.gov/docs/teachers/blackholes/blackholes.html>

- Power point -life cycles of stars.

<http://imagine.gsfc.nasa.gov/docs/teachers/lifecycles/stars.html>

You may find others by using Metacrawler (<http://www.metacrawler.com/>) and doing a global search on "life cycles of stars".

Questions

1. (a) What is a *nebula*?

- (b) What happens in a nebula?

2. How do *protostars* form?

3. There are 7 *main types of stars*. What are they and how are they classified?

Types:

Classification:

4. (a) What does the Hertzsprung-Russell diagram show about stars?

(b) What is plotted on this diagram? (Hint: What labels appear on the axes?)

5. (a) What is a "main sequence star" according to this diagram?

(b) Is our Sun a member of this group?

6. How are stars grouped according to their size? (There are *three groups* - think of their possible life cycles.)

(i)

(ii)

(iii)

7. Which stars become:

(a) black dwarfs?

(b) neutron stars?

(c) black holes?

8. What are the characteristics of the three stars listed in Q 7?

Black dwarf:

Neutron star:

Black hole:

9. (a) What will our star become when it is near the end of its life?

(b) Describe what will happen to our Sun in the next 5 billion years - use dot points.

10. (a) What is the difference between a *nova* and a *supernova*?
- (b) Which stars will supernova?
11. Find a *one-line description* for the following types of stars.
- (a) Yellow dwarf:
- (b) Red dwarf:
- (c) White dwarf:
- (d) Brown dwarf:
- (e) Red giant:
- (f) Blue giant:
- (g) Supergiant:
- (h) Pulsar:
12. (a) What is a quasar?
- (b) Is it a star?